

PCA plot showing the first two principal components (PC1 and PC2) of 18 morphological and physiological traits for 25 tree species. The x-axis represents PC1 (31.9%) and the y-axis represents PC2 (29.1%). Vectors indicate the direction of increasing trait values. Species are colored by family: blue (Fagaceae), green (Rosaceae), orange (Sapotaceae), and grey (Fabaceae).

Key traits and their associated species groups:

- Leaf Vcmax, Stomatal conductance, Leaf K:** Associated with Fagaceae (blue circles).
- Bark thickness, Stem diameter:** Associated with Rosaceae (green circles).
- Leaf thickness, Tree height, Wood density, Leaf density:** Associated with Sapotaceae (orange circles).
- Conduit diameter, Leaf N, Conduit diameter:** Associated with Fabaceae (grey circles).

PCA plot showing the relationship between morphological and physiological traits for 40 tree species. The x-axis represents PC1 (36.5%) and the y-axis represents PC2 (19.2%). Vectors indicate the direction of increasing values for various traits. Species are colored by life history: blue (evergreen), orange (deciduous), green (semi-deciduous), and triangles (pioneer).

Key traits and their associated species groups:

- Leaf K, Stomatal conductance, Bark thickness, Tree height, Leaf thickness:** Evergreen species (blue circles) cluster in the upper right quadrant.
- Leaf density, Wood density:** Evergreen species (blue circles) cluster in the lower right quadrant.
- Leaf area, Conduct diameter, Stem diameter, Leaf P, Specific leaf area, Leaf Vcmax:** Deciduous species (orange circles) cluster in the upper left quadrant.
- Leaf N, Leaf area, Conduct diameter, Stem diameter, Leaf P, Specific leaf area, Leaf Vcmax:** Semi-deciduous species (green circles) cluster in the lower left quadrant.
- Crown height, Leaf N, Leaf area, Conduct diameter, Stem diameter, Leaf P, Specific leaf area, Leaf Vcmax:** Pioneer species (triangles) cluster in the lower left quadrant.

N-fixing status $\triangle$ N-fixer $\bigcirc$ Non-fixer